

PRAKTIKUM SISTEM OPERASI

MODUL 9

Syscall



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. Soal 1

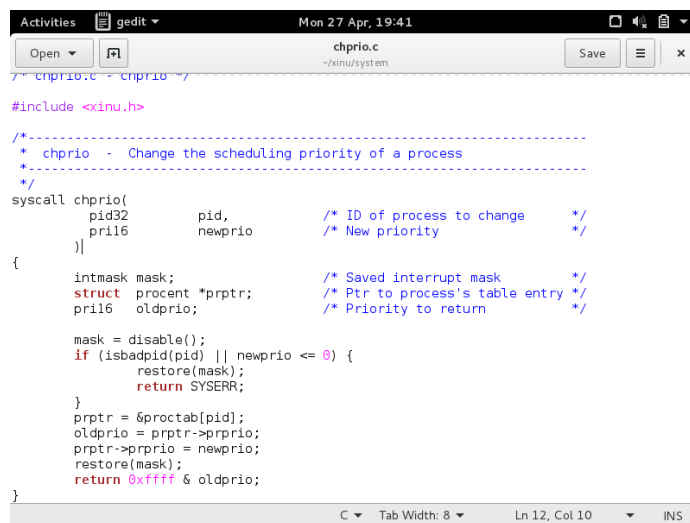
1. **[50 Poin]** Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan **Screenshot** kode dan hasil run)

2. Soal 2

2. **[25 Poin]** Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input
 - Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
 - Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make clean
- make



```
Activities gedit Mon 27 Apr, 19:41
chprio.c
~/xinu/system

/* chprio.c - chprio */

#include <xinu.h>

/*-----
 * chprio - Change the scheduling priority of a process
 *-----
 */

syscall chprio(
    pid32 pid, /* ID of process to change */
    pri16 newprio /* New priority */
)
{
    intmask mask; /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */
    pri16 oldprio; /* Priority to return */

    mask = disable();
    if (isbadpid(pid) || newprio <= 0) {
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->pprio;
    prptr->pprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

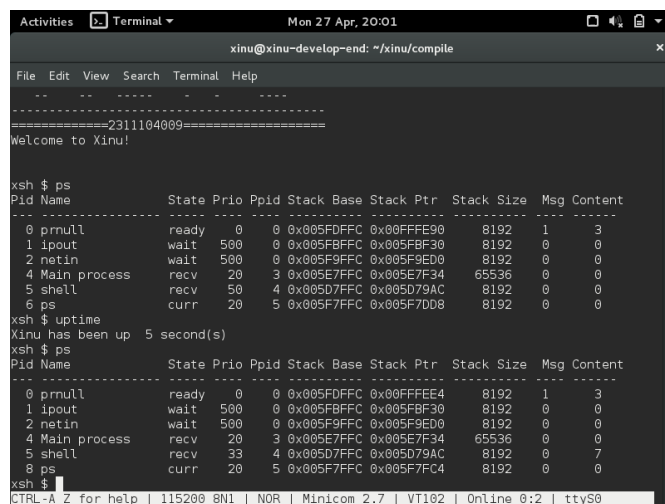
3. Soal nomer 3

3. Lakukan hal-hal berikut ini

- Edit xsh_uptime.c
Tambahkan kode berikut

```
/* Check for valid number of arguments */  
  
if (nargs > 1) {  
    fprintf(stderr, "%s: too many arguments\n", args[0]);  
    fprintf(stderr, "Try '%s --help' for more information\n",  
            args[0]);  
    return 1;  
}  
  
// tambahan code  
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah
 - make clean
 - make
- Jalankan perintah ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5
- Jalankan uptime
 - xsh \$ uptime
 - Perhatikan hasil perintah tersebut
- Jalankan ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah



```
Mon 27 Apr, 20:01  
xinu@xinu-develop-end: ~/xinu/compile  
-----  
2311104009-----  
Welcome to Xinu!  
  
xsh $ ps  
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size  Msg Content  
-----  
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE90 8192 1 3  
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0  
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0  
4 Main process rcv    20   3 0x005E7FFC 0x005E7F34 65536 0 0  
5 shell       rcv    50   4 0x005D7FFC 0x005D79AC 8192 0 0  
6 ps         curr   20   5 0x005F7FFC 0x005F7D08 8192 0 0  
  
xsh $ uptime  
Xinu has been up 5 second(s)  
xsh $ ps  
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size  Msg Content  
-----  
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3  
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0  
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0  
4 Main process rcv    20   3 0x005E7FFC 0x005E7F34 65536 0 0  
5 shell       rcv    33   4 0x005D7FFC 0x005D79AC 8192 0 7  
8 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0  
  
xsh $  
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

4. Soal nomor 4

[25 Poin] Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0 : Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c
- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”
- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)

```
Activities Terminal Mon 27 Apr, 20:08
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

-----2311104009-----
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps         curr   20  5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $ uptime
Xinu has been up 6 second(s)
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps         curr   20  5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

```
Activities Terminal Mon 27 Apr, 20:10
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu/shell x
^[[A^[[Anot found[[
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE84 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192 0 8
9 ps         curr   20  5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $ uptime
Xinu has been up 1 minute(s) 27 second(s)
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192 0 10
11 ps        curr   20  5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

